8.2 Practice: ratio and scale

← Unit overview · Answers

These sets start easy and build up. Use a ratio table (*verhoudingstabel*) whenever you like. There is no clock. If a question feels big, write down what belongs together first.

Set A: Ratio tables (Warm-up)

1. One pack has 4 yoghurts. How many yoghurts are in 3 packs?
2. 2 kg of apples costs €3. What does 4 kg cost?
3. 5 pens cost €2. What do 10 pens cost? And 15 pens?
4. A pancake recipe *voor 4 personen* needs 250 g flour (*bloem*), 500 ml milk and 2 eggs. Fill in the table.

personen	4	8	2	6
bloem (g)	250			
melk (ml)	500			
eieren	2			

5. 3 notebooks cost €4,50. What does 1 notebook cost? What do 7 cost?
6. Lemonade: 1 part syrup to 3 parts water. You use 50 ml syrup. How much water do you need? How much lemonade do you have in total?
7. In a class, 2 out of every 5 children have a pet. The class has 30 children. How many have a pet?
8. A bag holds 10 sweets. How many sweets are in 2 bags?

Talk about it: In number 4, how did you get the column for 6 people?

Set B: Scale on maps and plans (Building)

Tip: change cm into metres first. 100 cm = 1 m, 100 000 cm = 1 km.

1. A floor plan has scale 1 : 100. A wall on the plan is 5 cm. How long is the real wall?
2. A map of a park has scale 1 : 1 000. A path on the map is 3 cm. How long is the real path?
3. A walking map has scale 1 : 25 000. How many metres is 1 cm? How far is 4 cm in real life?
4. On a 1 : 25 000 map, a bike route is 10 cm. How many km is the real route?
5. On a 1 : 50 000 map, two villages are 6 cm apart. How far apart are they in real life?
6. On a 1 : 100 000 road map, the route to Oma's house is 7,5 cm. How many km is that?
7. You draw your bedroom at 1 : 50. The room is 4 m long. How long is it on your drawing?
8. On a 1 : 25 000 map, how many cm is a real distance of 2 km?
9. A scale line (*schaallijn*) shows that 2 cm on the map is 1 km. Write this as a scale 1 : …
10. A model bike is made at scale 1 : 10. The real bike is 1,80 m long. How long is the model, in cm?

Talk about it: Which question did you need the "change to metres" tip for?

Set C: Speed in km/u (Building → Stretch)

1. A car drives 60 km/u for 2 hours. How far does it go?
2. Femke cycles at 15 km/u. How far does she cycle in 1 hour? In half an hour?
3. You walk at 5 km/u for 3 hours. How far do you walk?
4. A train goes 120 km in 1 hour. How far does it go in 30 minutes?
5. A car drives 100 km/u. How long does it take to drive 250 km? Give your answer in hours and minutes.
6. Daan cycles 12 km in 30 minutes. What is his speed in km/u?
7. Sara walks 2 km in 20 minutes. What is her speed in km/u?
8. A train travels 40 km in 30 minutes. What is its speed in km/u?
9. You cycle at 18 km/u for 20 minutes. How far do you get?
10. A scooter (*brommer*) drives 45 km/u for 1 hour and 20 minutes. How far does it go?

Talk about it: Draw the ratio table you used for number 9. Which column did you make first?

Set D: Recipes and the best buy (Stretch → finish on a win)

1. Milk: 1 litre for €1,20, or 2 litres for €2,00. Which is the best buy? Show the price per litre.
2. Eggs: a box of 6 for €2,40, or a box of 10 for €3,50. Which is cheaper per egg?
3. Crisps (*chips*): 200 g for €1,50, or 300 g for €2,10. Which is cheaper per 100 g?
4. Hagelslag: 400 g for €2,40, or 600 g for €3,30. Which is cheaper per 100 g?
5. Shampoo costs €6 a bottle. Offer: *2e halve prijs* (second one half price). What do you pay for two bottles? What is that per bottle?
6. A pasta recipe *voor 4 personen* uses 300 g pasta. How much pasta do you need for 6 people?
7. A cookie recipe makes 20 cookies with 150 g butter. How much butter for 30 cookies?
8. Lemonade: 1 part syrup to 4 parts water. You want exactly 1 litre of lemonade. How many ml syrup and how many ml water?
9. Potatoes: a 5 kg bag for €4,50, or a 2 kg bag for €2,00. Which is cheaper per kilo?
10. In a class the ratio of boys to girls is 2 : 3. There are 25 children. How many boys and how many girls?
11. A recipe for 2 people uses 1 egg. How many eggs for 4 people?

Talk about it: In number 9, is the cheapest-per-kilo bag always the best choice? When not?

Puzzle corner (optional, just for fun)

1. A snail crawls at 5 metres per hour. How many days would it take to crawl 1 km, if it never stopped?
2. On a 1 : 25 000 map, a square field is 2 cm by 2 cm. How long is each side in real life? (Bonus: the area of the real field is not “4 times” the map area. How many times bigger is it?)
3. Two candles of the same length: one burns down in 4 hours, the other in 6 hours. You light both at the same time. After how long is the first one exactly half as long as the second?